TECHNICAL SHEET · PROJECT

NEXUS

Research prototype: skin and eye image analysis, contactless heart-rate measurement and federated learning, on the web, on mobile and on a Raspberry Pi 5.

> **NEXUS is not a medical device**
> It is a research prototype built as a student project. It is not certified and has not been clinically validated. It gives no diagnosis, no medical advice and no reliable screening, and its results can be wrong. It must not be used to make a health decision. If you have a concern about your health, see a healthcare professional.

≈ 95.9%
eye-model accuracy, 5 classes, 344 evaluation images

61.6%
skin-model accuracy, 11 classes (a harder task)

48 ms
per eye inference on Raspberry Pi 5, mean of 30 runs

3
platforms sharing the same ONNX files: web, mobile, edge

1. What the system does

The same inference chain runs locally on three platforms: a web app (Streamlit), a mobile app (Flutter, Android) and a Raspberry Pi 5. A federated learning server (Flower, FedAvg) aggregates model weights. Raw images never leave the device that captured them.

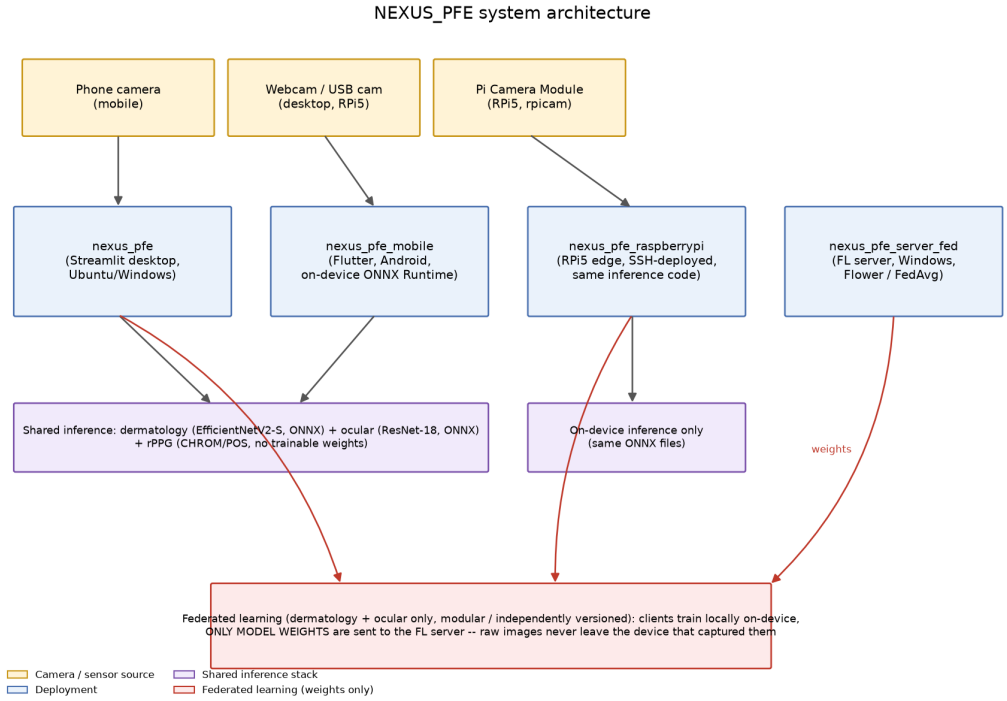


Figure 1. Overall architecture. Source: the project's public repository.

2. Models and results

Module	Model	Classes	Result	Notes
Eye	ResNet-18 (PyTorch) → ONNX	5	≈ 95.9% accuracy	Per-class F1 ≥ 0.93. Confusion matrix over 344 images. The repository notes it is not yet confirmed whether this is a true held-out test set or a validation split.
Skin	EfficientNetV2-S (TensorFlow/Keras) → ONNX	11	61.6% accuracy	Precision 0.79 and recall 0.46 (macro). Weakest class: psoriasis (F1 0.39). 42 epochs, fine-tuning from epoch 31.
rPPG	CHROM and POS, signal processing, no trained weights	–	73.4 BPM recovered for 72 BPM injected	Measured on a synthetic clip only. Not yet compared with a pulse oximeter on a real person.

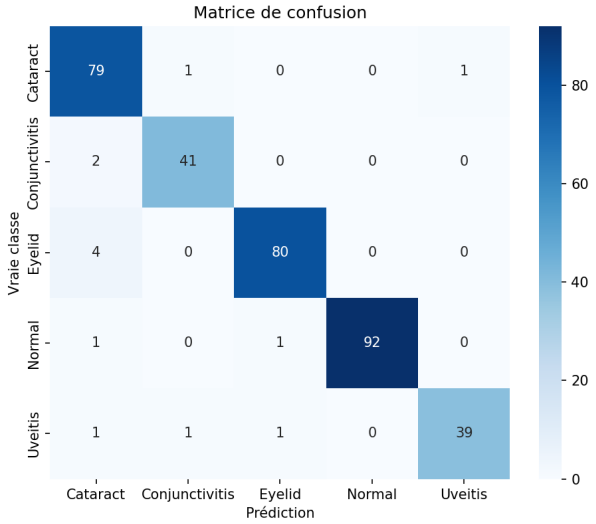


Figure 2. Confusion matrix, eye model (5 classes).

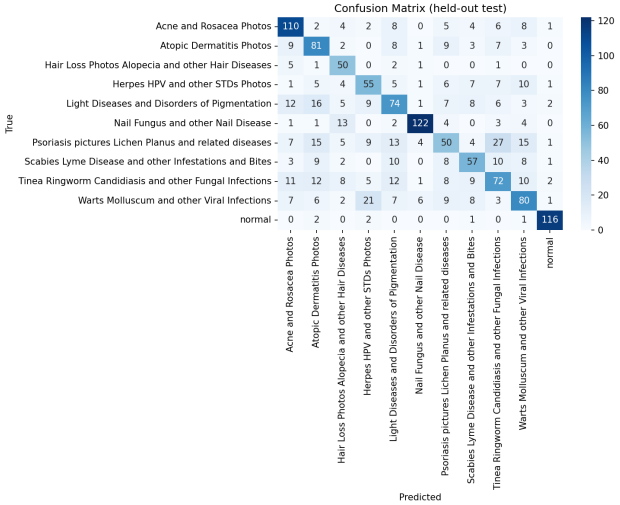


Figure 3. Confusion matrix, skin model (11 classes).

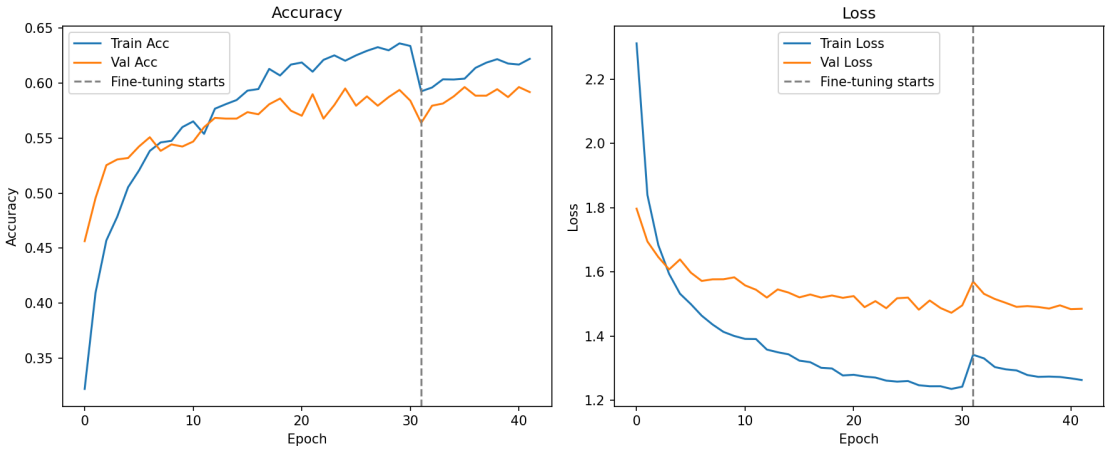


Figure 4. Learning curves of the skin model.

3. Measurements on the three platforms

30 timed runs per platform, after 5 discarded warm-up runs.

Platform	Skin latency	Eye latency	Skin memory	Eye memory	Skin load	Eye load
Web (Streamlit)	46 ms	16 ms	229 MB	173 MB	802 ms	135 ms
Mobile (Flutter)	386 ms	245 ms	487 MB	442 MB	3,143 ms	409 ms
Raspberry Pi 5	105 ms	48 ms	250 MB	229 MB	1,263 ms	39 ms

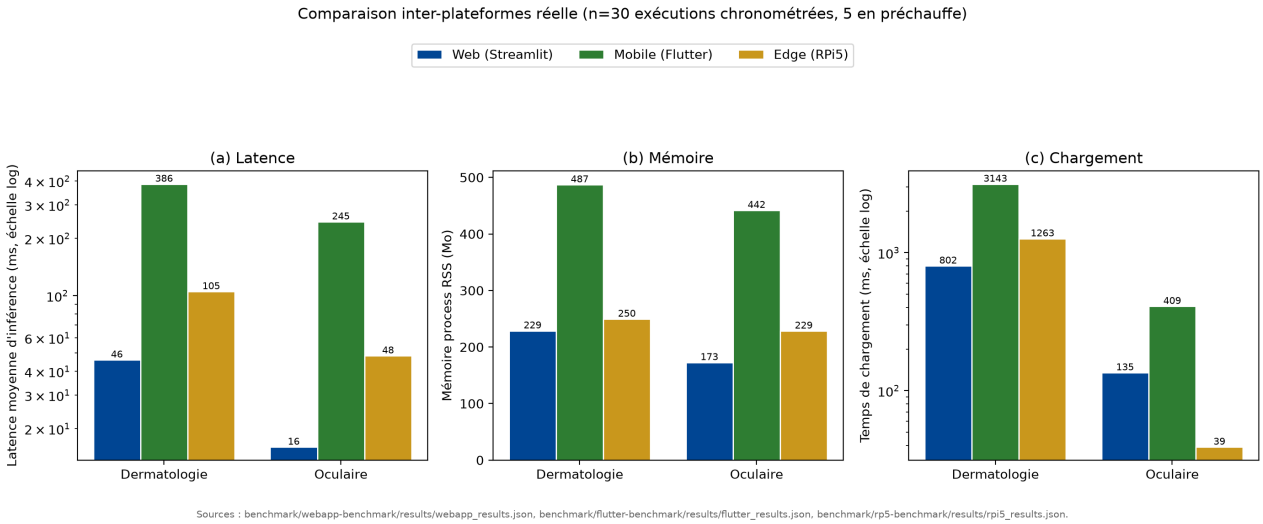


Figure 5. Latency, memory and load time on the three platforms (figure labels are in French).

Of 6 images tested identically on the three platforms, 5 give exactly the same prediction. The one mismatch is an error of the model itself, not a porting defect.

4. Federated learning

Two clients train locally and send only weights to the server (FedAvg). Over 3 rounds, measured on held-out real data, the skin model stays around 59% and the eye model between 99.5% and 99.7%. Performance therefore holds in a federated setting. On mobile, training does not run on the device: the screen shows the server's real history.

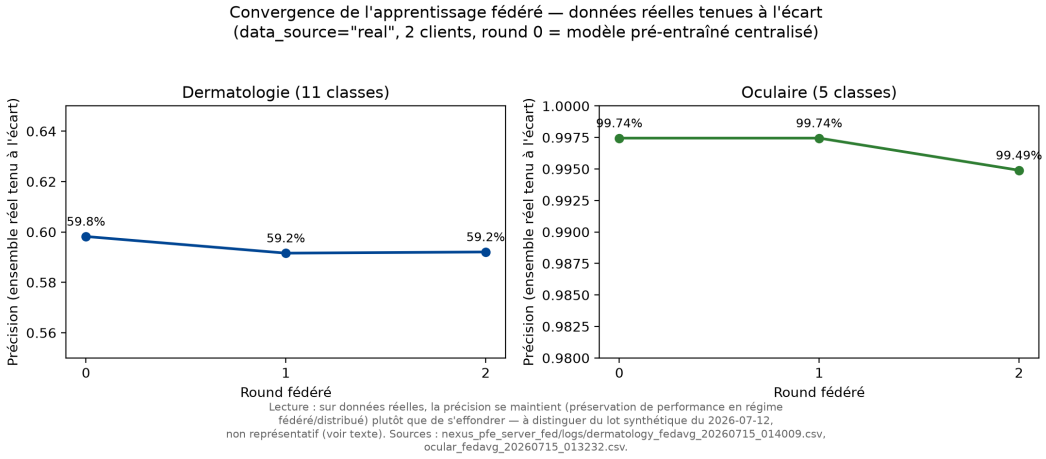


Figure 6. Federated learning convergence on real data (figure labels are in French).

5. Screenshots

Screenshots from the public repository. The input images are blurred there on purpose.

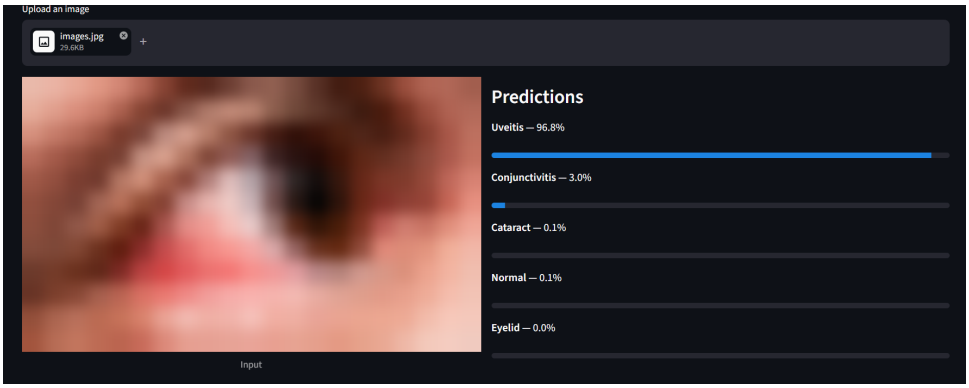


Figure 7. Web app, eye tab: uploaded image and per-class scores.

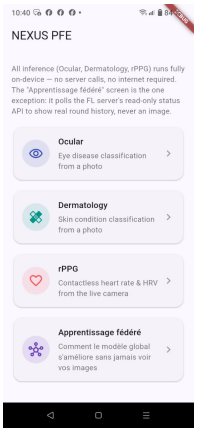


Figure 8. Mobile: home.

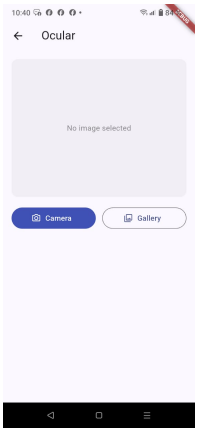


Figure 9. Mobile: eye module.

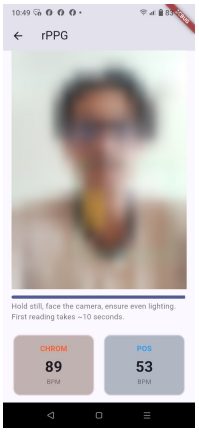


Figure 10. Mobile: rPPG, CHROM and POS estimates.

6. Hardware

Item	Role
Raspberry Pi 5	Edge deployment: same inference code and same ONNX files as the web app.
Cameras	USB webcam, IP camera or the Raspberry Pi camera module. MediaPipe face detection with automatic fallback to OpenCV (Haar cascade).
Android phone	Flutter app with on-device ONNX Runtime, no server call for inference.

The repository does not yet contain a photo of the hardware setup.

7. Known limitations

- The skin model is still unreliable (61.6%): eleven visually similar conditions.
- rPPG has not yet been compared with a reference measurement on a real person.
- According to its authors, the eye dataset comes from online images balanced with SMOTE: the reported accuracy should be read with caution.
- No clinical validation, no certification.

8. Sources and credits

All numbers and figures come from the public repository: github.com/yassinefartassi/nexus-health-screening (README, docs/model_cards.md, figures/). Datasets: DermNet (dermnetnz.org, Kaggle copy by S. Goel); M. Ahmed, Image Dataset on Eye Diseases Classification, Mendeley Data, 2024. rPPG algorithms adapted from rPPG-Toolbox (de Haan and Jeanne, 2013; Wang et al., 2017). Federated learning: McMahan et al., 2017, with Flower. Neither the datasets nor the model weights are published.